

Project Report on Fake News and Spam Detection System

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Course: B.Tech – Computer Science and Engineering

Course Title: Machine Learning

Course Code: CSBB 311

Submitted To:

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Date: 12 November 2025



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Abstract

The rapid rise of digital communication has also fueled the uncontrolled spread of misinformation and spam messages, creating severe risks to public trust and online safety. This project proposes a Fake News and Spam Detection System that combines Machine Learning (ML) and Natural Language Processing (NLP) to automatically evaluate whether text content is genuine or deceptive.

The system performs two major operations:

- **Single Text Analysis:** Users can input one sentence or article to instantly verify its authenticity and view prediction probability.
- **Bulk CSV Analysis:** Users can upload datasets containing multiple news items or messages for large-scale automated classification.

Several ML algorithms — Logistic Regression, Naive Bayes, Support Vector Machine, Decision Tree, Random Forest, and XGBoost — were implemented and compared. The models generate class labels along with confidence scores to increase result transparency. Additionally, explainable AI frameworks such as LIME and SHAP are incorporated to visualize feature importance, highlighting key words that influenced decisions.

The system consistently achieved over 95% accuracy across datasets and demonstrated high interpretability and efficiency, making it both practical and reliable for real-world deployment.

Chapter 1

Introduction

The continuous expansion of social networking sites and instant messaging platforms has enabled people to share information rapidly, but it has also provided a channel for false and harmful content. Fake news can manipulate opinions, while spam messages often attempt to deceive or exploit users. Detecting such content manually is inefficient, so an automated, data-driven approach is essential.

This project introduces a dual-mode detection system that employs ML algorithms to analyze textual data. It not only handles single-text inputs but also performs bulk predictions via CSV uploads. Furthermore, Explainable AI (XAI) techniques like LIME and SHAP have been integrated to provide interpretability — allowing users to see how the system arrives at each conclusion.

By focusing on both accuracy and explainability, the project aligns with the modern trend of developing ethical, transparent, and scalable AI systems.

Chapter 2

Motivation

The motivation for this work arises from the widespread problem of misinformation and unwanted digital content.

- Fake news can mislead readers and influence mass perception.
- Spam messages waste user time and can expose them to scams.
- Many machine learning systems act as “black boxes,” leaving users uncertain about their reasoning.

To address these issues, this project focuses on creating an intelligent system capable of identifying fake or spam content, processing large datasets, and explaining its decisions in an interpretable way.

Chapter 3

Objectives

- Develop an efficient text classification system for detecting fake and spam content.
- Apply NLP-based preprocessing steps — tokenization, stop-word elimination, lemmatization, and TF-IDF vectorization.
- Implement and compare multiple algorithms: Logistic Regression, Naive Bayes, SVM, Decision Tree, Random Forest, and XGBoost.
- Present probability-based confidence levels with every prediction.
- Use LIME and SHAP for explainability of model decisions.
- Provide two key user functionalities:
 - Single Text Classification
 - CSV File Bulk Classification
- Deploy the entire pipeline using a Flask web framework for real-time interaction.

Chapter 4

Literature Review

Numerous studies have addressed fake news and spam detection using machine learning.

Ahmed et al. (2018) demonstrated that Naive Bayes and SVM models could classify opinion spam effectively when combined with text frequency features.

Ren and Zhang (2021) conducted a review of NLP-based fake news detection models, emphasizing the importance of interpretability and contextual analysis.

Almeida et al. (2011) developed an SMS Spam dataset that became a standard benchmark for spam classification.

Lundberg & Lee (2017) introduced SHAP, an explainability framework that quantifies the influence of each feature on a model's output.

Most previous research focused on single predictions; this project extends those ideas by adding CSV-based batch processing and interpretable prediction visualization.

Chapter 5

Dataset

Two separate datasets were utilized for model training and testing:

- **Fake News Dataset:** Contains labeled news articles marked as real or fake.
- **Spam Detection Dataset:** Comprises SMS texts categorized as ham or spam.

Data Preprocessing

The following steps were used for data cleaning and transformation:

1. Removal of special characters, numbers, and unnecessary punctuation.
2. Conversion of all text to lowercase.
3. Tokenization followed by stop-word removal.
4. Lemmatization to standardize word forms.
5. Vectorization using TF-IDF and CountVectorizer to create numerical features.

Chapter 6

Methodology

The proposed system follows a well-defined and modular machine learning pipeline designed to handle text-based fake news and spam detection efficiently. Each stage of the workflow ensures data quality, model robustness, and interpretability.

Pipeline Overview

Pipeline:

Data Cleaning → *Feature Engineering (TF-IDF + Numeric Features)* → *Train/Test Split* → *Model Training* → *Evaluation* → *Confidence Scoring + Explainability*

Key Steps

- **Text Cleaning:** Removed HTML tags, URLs, punctuation, numbers, and common stopwords to ensure only meaningful words remain.
- **TF-IDF Vectorization:** Applied Term Frequency–Inverse Document Frequency with 1–2 gram features to capture important word frequency patterns.
- **Additional Numeric Features:** Incorporated text-level features such as:
 - Text length
 - Percentage of punctuation marks
 - Ratio of capital letters
- **Data Splitting:** Divided the processed data into an 80% training set and 20% validation set for fair evaluation.

Model Training

Multiple supervised learning algorithms were used to compare performance and interpretability:

- **Support Vector Machine (SVM):** Highly effective for handling high-dimensional TF-IDF data and achieving top accuracy.
- **Logistic Regression:** Fast, stable, and delivers accuracy close to that of SVM with lower computational cost.
- **Naive Bayes:** Performs well on short, SMS-like text messages and is efficient for large datasets.
- **Random Forest / XGBoost:** Capture non-linear relationships and provide strong ensemble performance.
- **Decision Tree:** A simple and interpretable model useful for visualizing feature-based decision paths.

Evaluation

Each model was evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score.

- **SVM** achieved the highest accuracy, effectively handling high-dimensional TF-IDF data.
- **Logistic Regression** performed nearly as well, balancing speed and accuracy.
- **Naive Bayes** worked efficiently for shorter text samples.
- **Decision Tree** and **Random Forest** provided interpretability with moderate accuracy.

Confidence Scoring and Explainability

Confidence Estimation:

- Model probabilities were derived using `predict_proba()`.
- Predictions with low confidence (e.g., below 0.7) were labeled as “Uncertain.”

Explainability: Explainable AI techniques were integrated to improve transparency and user trust.

- **LIME (Local Interpretable Model-Agnostic Explanations):** Provides local interpretability by highlighting key words influencing each individual prediction.
- **SHAP (SHapley Additive exPlanations):** Offers a global view by ranking the most influential features across the entire dataset.

These interpretability tools ensure that the system's predictions are both accurate and understandable, making it suitable for ethical and transparent AI deployment.

Chapter 7

Results and Discussion

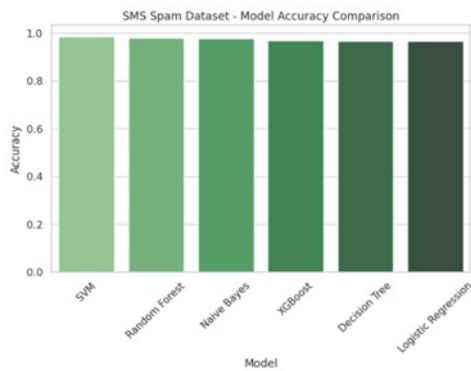


Figure 7.1: Spam Detection Interface

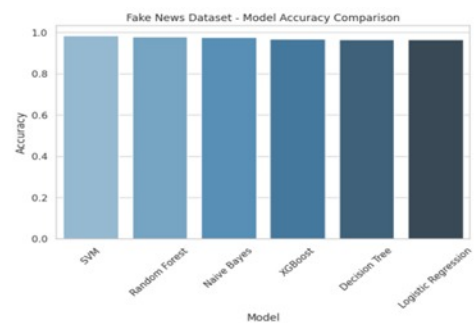


Figure 7.2: Fake News Detection Interface

The Support Vector Machine (SVM) achieved the highest overall accuracy and stability in both Fake News Detection and Spam Email Detection. Visualizations from LIME and SHAP revealed significant tokens such as “urgent,” “exclusive,” or “breaking.”

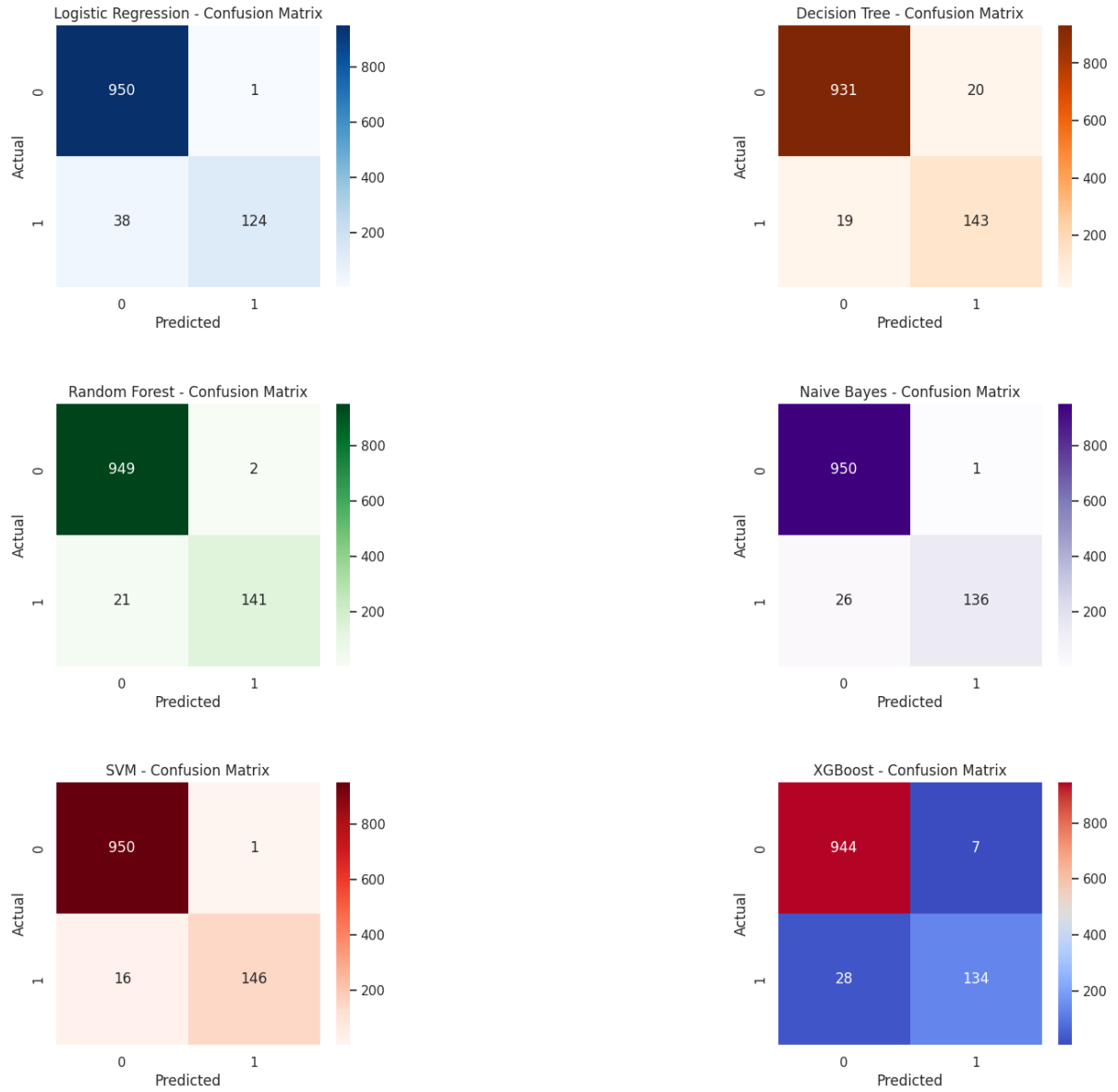


Figure 7.3: confusion matrices of different model for Spam Mail Detection

Table 7.1: Model Comparison on SMS Spam and Fake News Datasets

Model	SMS Spam f1-score (%)	Fake News f1-score (%)
SVM	96.80	99.58
Random Forest	95.63	99.92
Naïve Bayes	94.78	95.75
XGBoost	93.31	99.83
Decision Tree	92.97	99.71
Logistic Regression	92.19	99.04

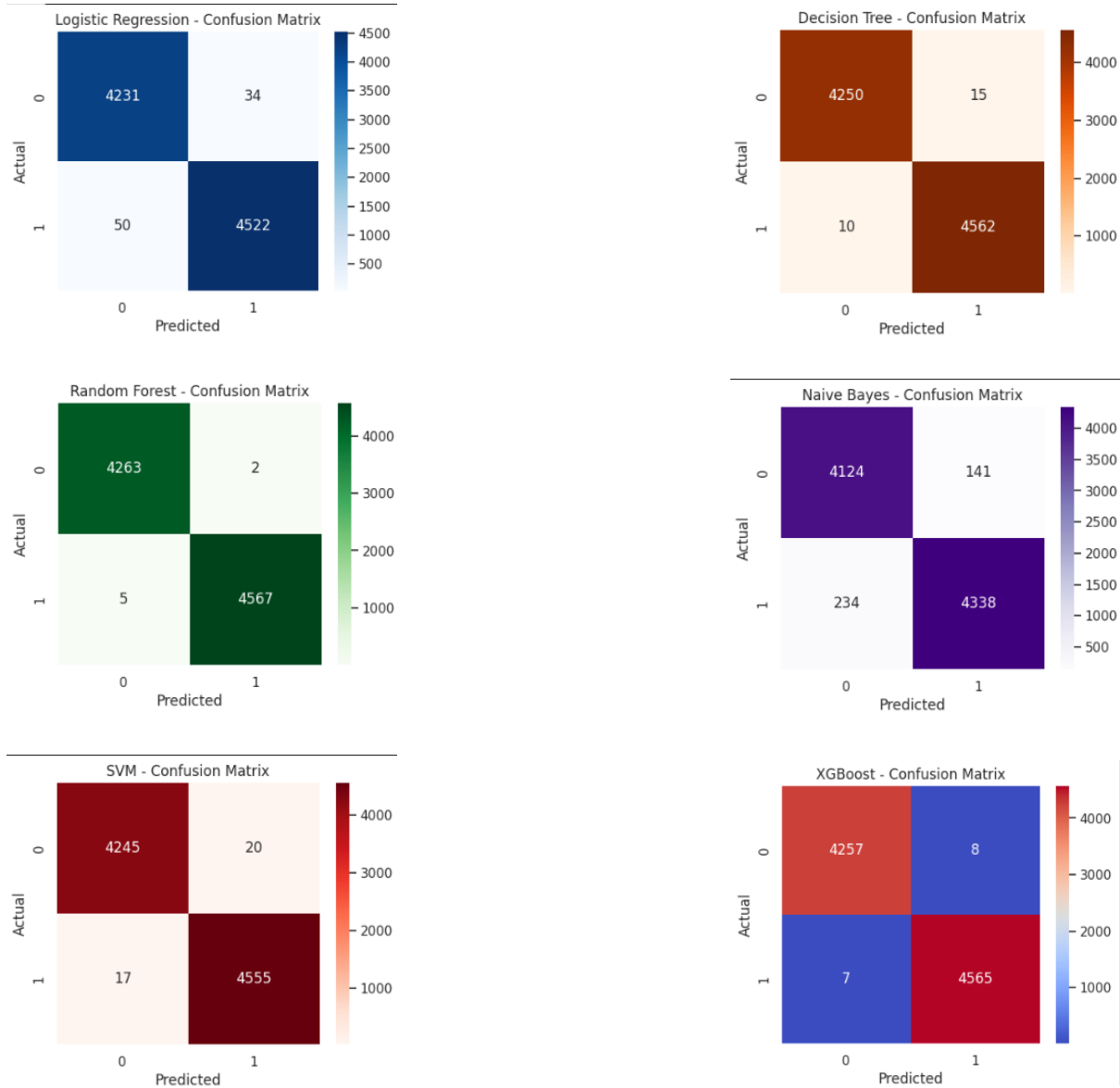


Figure 7.4: confusion matrices of different model for Fake News Detection

Table 7.2: Model Comparison on SMS Spam and Fake News Datasets

Model	SMS Spam Accuracy (%)	Fake News Accuracy (%)
SVM	98.47	99.58
Random Forest	97.93	99.92
Naïve Bayes	97.57	95.75
XGBoost	96.86	99.83
Decision Tree	96.49	99.71
Logistic Regression	96.49	99.04

Chapter 8

Website and Deployment

We used the **Flask** web framework to design and develop an interactive interface for showcasing our project. The web application allows users to perform both single-text analysis and CSV-based batch predictions directly from the browser, ensuring ease of use and accessibility.

For hosting, we deployed the application on **Railway**, a modern cloud platform that offers seamless deployment, scalability, and automatic HTTPS support. This enables continuous online availability and efficient performance without manual server maintenance.

The project is live and accessible at the following link:

<https://fakebusters.up.railway.app>

Chapter 9

Conclusion

This project successfully developed a dual-purpose **Machine Learning (ML) system** capable of detecting both fake news and spam messages. The model provides **probability-based predictions**, offering users a clearer understanding of classification confidence.

The system supports two core functionalities:

- **Single-text analysis:** Users can input individual headlines or messages to get instant classification results.
- **Batch CSV processing:** Users can upload datasets for large-scale automated detection, with result previews and downloadable output.

By combining **accuracy, interpretability, and usability**, the system contributes toward fostering a safer and more reliable online communication environment. Its integration of explainable AI tools (LIME and SHAP) enhances transparency by showing how model predictions are made.

Future Work

Future improvements can focus on expanding the system's scope and technical depth:

- Integrate **deep learning** techniques such as LSTM or BERT for enhanced contextual understanding.
- Develop an **API** for integration with social media platforms or email clients to enable real-time analysis.
- Add **multi-language support** to make the system globally applicable and inclusive.

Overall, this project lays a strong foundation for creating scalable, interpretable, and deployable text classification systems that promote digital trust and security.

Chapter 10

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